

THEORY OF COMPUTATION (CS302)

DFA and NFA Diagrams

Learning Objectives

Understand Finite Automata, Deterministic Finite Automata (DFA), Non-Deterministic Finite Automata (NFA), transition diagrams, transition tables, and their applications.

Introduction to Finite Automata

Finite Automata are mathematical models used to recognize patterns and formal languages. They form the foundation of compiler design, lexical analysis, and text processing systems.

Deterministic Finite Automata (DFA)

A DFA is a finite state machine in which for every state and input symbol there is exactly one next state.

Components of DFA

- Finite set of states (Q)
- Input alphabet (Σ)
- Transition function (δ)
- Start state (q_0)
- Set of final states (F)

Example DFA

Language: Strings ending with '1'

$q_0 \xrightarrow{1} q_1$
 $q_0 \xrightarrow{0} q_0$
 $q_1 \xrightarrow{0} q_0$
 $q_1 \xrightarrow{1} q_1$

Transition Table for DFA

State	0	1
q_0	q_0	q_1
q_1	q_0	q_1

Non-Deterministic Finite Automata (NFA)

An NFA is a finite state machine where multiple transitions for the same input symbol may exist from a state.

Components of NFA

- Finite set of states
- Input alphabet
- Transition function
- Initial state
- Final states

Example NFA

$q_0 \xrightarrow{0} q_0$
 $q_0 \xrightarrow{0} q_1$
 $q_1 \xrightarrow{1} q_2$

Transition Table for NFA

State | 0 | 1
 q_0 | $\{q_0, q_1\}$ | $\emptyset$
 q_1 | $\emptyset$ | q_2
 q_2 | $\emptyset$ | $\emptyset$

DFA vs NFA

DFA: Exactly one transition per symbol.

NFA: Multiple transitions allowed.

Both accept the same class of regular languages.

Applications

Finite Automata are used in lexical analyzers, pattern matching, network protocol design, digital circuits, and text processing systems.

Exam Important Questions

1. Define DFA.
2. Define NFA.
3. Explain components of DFA.
4. Compare DFA and NFA.
5. Construct DFA for a given language.
6. Convert NFA to DFA.

Summary

DFA and NFA are fundamental automata models used to recognize regular languages. Understanding their structure, diagrams, and transition functions is essential in Theory of Computation.